ID No: __________________Name: ________________________________Sec No.: _______

**BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI**

**II SEMESTER 2025-26**

**A**

**CS/ECE/EEE/INSTR F241 Microprocessor Programming & Interfacing**

**28-01-2026 (Wednesday)** **QUIZ-1 (closed book)**

**Max time: 15 Min** **Max Marks: 08**

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Q1. A memory device is **nibble (4-bit) organized** and has a total storage capacity of **8 MB**. Determine the **number of address lines** required for this memory device. **[2M]**

**Show the calculations below:**

| **Calculation** | **Final Answer** |
|---|---|
| | **Answer: 24** |

**Q2.** Consider the following 8086 assembly instruction: ADD AX, BX. Assume that condition of all the status flags of the flag register are cleared (RESET) before the execution of the instruction.

AX = F200H

BX = 3200H

After executing the instruction, determine the **status (SET(1) or RESET (0))** of the following flags:

**(1) Carry Flag**

**(2) Overflow Flag** **[1M+1M]**

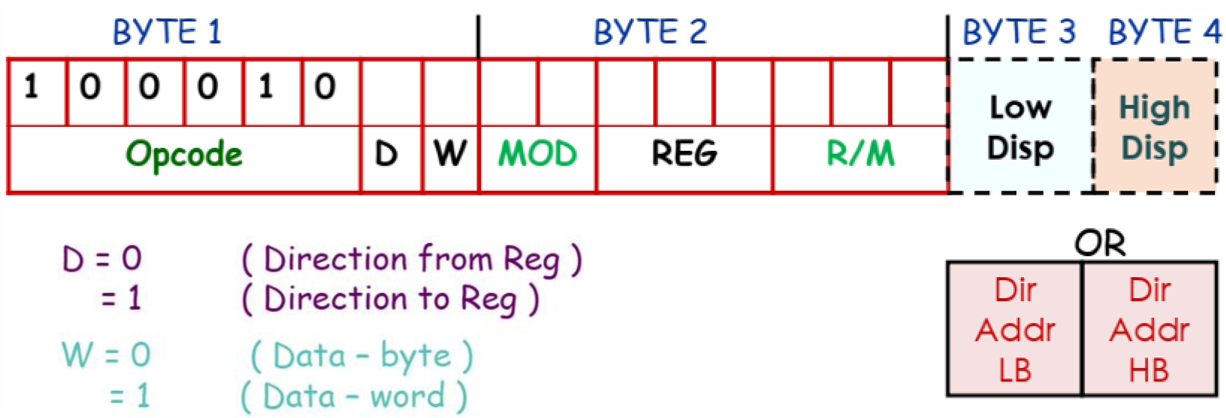
| **Calculation** | **Final Answer** |
|---|---|
| | **Carry Flag- 1, Overflow Flag=0** |

**Q3. For the 8086 processor, CS = $1F40_H$, DS = $IF20_H$, SS = $IF00_H$, BX = $0840_H$, SI = $0120_H$, DI = $2100_H$, DX = $1400_H$.** A memory access is made for address $1FF00_H$ in the Code Segment (CS), with a base-relative, indexed addressing mode**. A positive displacement is utilized**. SI is used as the index register. Mention the instruction. Only the registers mentioned above are used in the instruction. Show Calculation **[2M]**

| **1m if CS:[BX + SI + 01A0H] is mentioned** | **Final Answer**<br><br>**MOV DX, CS:[BX + SI + 01A0H]** |
|---|---|

**Q.4. Find the machine code for the instruction MOV DL, [BX+10H] for 8086 processors. Write the machine code in hexadecimal only. Show Calculation. [2M]**

| **Calculation** | **Final Answer (in hexadecimal)**<br><br>**8A 57 10** |
|---|---|




| MOD R/M | 00 | 01 | 10 | 11 W = 0 | 11 W = 1 |
|---|---|---|---|---|---|
| 000 | [BX] + [SI] | [BX]+[SI] +d8 | [BX]+[SI]+d16 | AL | AX |
| 001 | [BX] + [DI] | [BX]+[DI]+d8 | [BX]+[DI]+d16 | CL | CX |
| 010 | [BP] + [SI] | [BP]+[SI]+d8 | [BP]+[SI]+d16 | DL | DX |
| 011 | [BP] + [DI] | [BP]+[DI]+d8 | [BP]+[DI]+d16 | BL | BX |
| 100 | [SI] | [SI]+d8 | [SI]+d16 | AH | SP |
| 101 | [DI] | [DI]+d8 | [DI]+d16 | CH | BP |
| 110 | d16 | [BP] + d8 | [BP] + d16 | DH | SI |
| 111 | [BX] | [BX]+d8 | [BX]+d16 | BH | DI |

| MOD R/M | 00 | 01 | 10 | 11 W = 0 | 11 W = 1 |
|---|---|---|---|---|---|
| 000 | EAX | EAX+d8 | EAX+d32 | AL | EAX |
| 001 | ECX | ECX+d8 | ECX+d32 | CL | ECX |
| 010 | EDX | EDX+d8 | EDX+d32 | DL | EDX |
| 011 | EBX | EBX+d8 | EBX+d32 | BL | EBX |
| 100 | Scaled Index | Scaled Index +d8 | Scaled Index +d32 | AH | ESP |
| 101 | d32 | EBP+d8 | EBP+d32 | CH | EBP |
| 110 | ESI | ESI+d8 | ESI+d32 | DH | ESI |
| 111 | EDI | EDI+d8 | EDI+d32 | BH | EDI |

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B

**Max time: 15 Min** **Max Marks: 08**

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Q.1. A memory device is **nibble (4-bit) organized** and has a total storage capacity of 16 **MB**. Determine the **number of address lines** required for this memory device. **[2M]**

**Show the calculations below:**

| **Calculation** | **Final Answer**<br><br>Answer: 25 |
|---|---|

**Q2.** Consider the following 8086 assembly instruction: ADD AX, BX. Assume that condition of all the status flags of the flag register are cleared (RESET) before the execution of the instruction.

AX = F200H

BX = 3200H

After executing the instruction, determine the **status (SET(1) or RESET (0))** of the following flags:

**(1) Parity Flag**

**(2) Auxiliary Carry Flag**

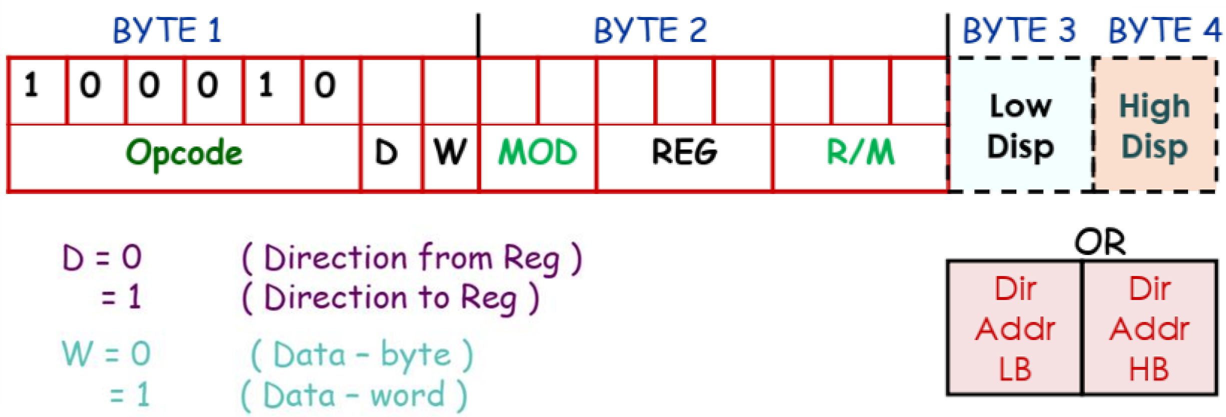
| **Calculation** | **Final Answer**<br><br>**PF=1**<br>**AF=0** |
|---|---|

**Q3.For the 8086 processor, CS = $1F40_H$, DS = $IF20_H$, SS = $IF00_H$, BX = $0840_H$, SI = $0120_H$, DI = $2100_H$, DX = $1400_H$.** A memory access is made for address $1FCC60_H$ in the Code Segment (CS), with a base-relative, indexed addressing mode**. A negative displacement is utilized**. SI is used as the index register. Mention the instruction. Only the registers mentioned above are used in the instruction. Show Calculation

| **1M If CS:[BX + SI - 4100H] is mentioned** | **Final Answer**<br><br>**MOV DX, CS:[BX + SI - 4100H]** |
|---|---|

**Q.4. Find the machine code for the instruction MOV  [BP+10H], DL for 8086 processors. Write the machine code in hexadecimal only. Show Calculation.  [2M]**

| **Calculation** | **Final Answer (in hexadecimal)**<br><br>**88 56 10** |
|---|---|




| MOD R/M | 00 | 01 | 10 | 11 W = 0 | 11 W = 1 |
|---|---|---|---|---|---|
| 000 | [BX] + [SI] | [BX]+[SI] +d8 | [BX]+[SI]+d16 | AL | AX |
| 001 | [BX] + [DI] | [BX]+[DI]+d8 | [BX]+[DI]+d16 | CL | CX |
| 010 | [BP] + [SI] | [BP]+[SI]+d8 | [BP]+[SI]+d16 | DL | DX |
| 011 | [BP] + [DI] | [BP]+[DI]+d8 | [BP]+[DI]+d16 | BL | BX |
| 100 | [SI] | [SI]+d8 | [SI]+d16 | AH | SP |
| 101 | [DI] | [DI]+d8 | [DI]+d16 | CH | BP |
| 110 | d16 | [BP] + d8 | [BP] + d16 | DH | SI |
| 111 | [BX] | [BX]+d8 | [BX]+d16 | BH | DI |

| MOD R/M | 00 | 01 | 10 | 11 W = 0 | 11 W = 1 |
|---|---|---|---|---|---|
| 000 | EAX | EAX+d8 | EAX+d32 | AL | EAX |
| 001 | ECX | ECX+d8 | ECX+d32 | CL | ECX |
| 010 | EDX | EDX+d8 | EDX+d32 | DL | EDX |
| 011 | EBX | EBX+d8 | EBX+d32 | BL | EBX |
| 100 | Scaled Index | Scaled Index +d8 | Scaled Index +d32 | AH | ESP |
| 101 | d32 | EBP+d8 | EBP+d32 | CH | EBP |
| 110 | ESI | ESI+d8 | ESI+d32 | DH | ESI |
| 111 | EDI | EDI+d8 | EDI+d32 | BH | EDI |